Circles

Class 3

1 Arcs

Two points A and B on a circle cut it into two **arcs**. The smaller one is the **minor arc** and the larger one is the **major arc**. If AB is a diameter, the two arcs are equal; each is a **semicircle**.

FIG

The angle that an arc **subtends at the centre** is the angle swept as the radius turns from OA to OB along that arc. The minor arc AXB subtends $\angle AOB$. The major arc AYB subtends the reflex angle AOB, which is $360° - \angle AOB$.

Solve first.

P1. In a circle with centre O, $\angle AOB = 100°$.

(a) Which arc subtends $100°$ at the centre: the minor arc or the major arc?

(b) What angle does the other arc subtend at the centre?

(c) What angle does a semicircle subtend at the centre?

2 The angle at the centre and the angle on the circle

The angle that arc AB **subtends at a point** C on the rest of the circle is $\angle ACB$. In P2 to P4, O is the centre and CD is a diameter. Draw a rough diagram for each and mark every angle you find.

Solve first.

P2. A is a point on the circle, and $\angle OCA = 25°$.

(a) Why is $\angle OAC = 25°$?

(b) $\angle AOD$ is an exterior angle of $\triangle OAC$. Find it.

(c) Compare $\angle AOD$ with $\angle ACD$.

P3. A and B lie on opposite sides of the diameter CD, with $\angle OCA = 20°$ and $\angle OCB = 30°$.

(a) Use P2 to find $\angle AOD$ and $\angle BOD$.

(b) Find $\angle AOB$ and $\angle ACB$.

(c) Compare $\angle AOB$ with $\angle ACB$.

P4. A and B lie on the same side of the diameter CD, with $\angle OCA = 40°$ and $\angle OCB = 15°$.

(a) Find $\angle AOD$ and $\angle BOD$.

(b) Find $\angle AOB$ and $\angle ACB$.

(c) Compare $\angle AOB$ with $\angle ACB$. What pattern do P2, P3 and P4 show?

Theorem 9 (NCERT Theorem 9). *The angle subtended by an arc at the centre of the circle is double the angle subtended by the arc at any point on the circle outside the arc.*

FIG

Why it works.

Given: arc AB of a circle with centre O; C is a point on the circle outside the arc.
To show: $\angle AOB = 2\angle ACB$.

1. Draw the diameter CD. $OA = OC$ (radii), so $\angle OAC = \angle OCA$. The exterior angle of $\triangle OAC$ is $\angle AOD = \angle OAC + \angle OCA = 2\angle OCA$.
2. In the same way, $\angle BOD = 2\angle OCB$.
3. If O lies inside $\angle ACB$, add: $\angle AOB = \angle AOD + \angle BOD = 2(\angle OCA + \angle OCB) = 2\angle ACB$.
4. If O lies outside $\angle ACB$, subtract: $\angle AOB = \angle AOD - \angle BOD = 2(\angle OCA - \angle OCB) = 2\angle ACB$.

The same steps work when the arc is a major arc; then $\angle AOB$ is a reflex angle.

Example 1. In a circle with centre O, $\angle AOB = 130°$, and C is a point on the major arc AB. Find $\angle ACB$ and $\angle OAB$.

FIG

Given: $\angle AOB = 130°$; C is on the major arc.
To find: $\angle ACB$ and $\angle OAB$.

1. C is outside the minor arc AB, which subtends $130°$ at O. So $\angle ACB = \dfrac{130°}{2} = 65°$ (Theorem 9).
2. $OA = OB$ (radii), so $\angle OAB = \angle OBA = \dfrac{180° - 130°}{2} = 25°$.

Answer: $\angle ACB = 65°$ and $\angle OAB = 25°$.

3 Two consequences

Solve first.

P5. In a circle with centre O, $\angle AOB = 80°$. C, D and E are three points on the major arc AB.

(a) Find $\angle ACB$, $\angle ADB$ and $\angle AEB$.
(b) Does the answer depend on where C is on the major arc?

P6. AB is a diameter of a circle with centre O, and C is any other point on the circle.

(a) What angle does the semicircle not containing C subtend at O?
(b) Find $\angle ACB$.

The part of the circle on one side of a chord, together with the chord, is a **segment**. P5 shows that *angles in the same segment are equal*: each is half the same angle at the centre.

Corollary of Theorem 9 (NCERT Corollary). *The angle subtended by a diameter at any point on the circle is* $90°$.

FIG

Why it works.

Given: AB is a diameter of a circle with centre O; C is any other point on the circle.
To show: $\angle ACB = 90°$.

1. The semicircle not containing C subtends a straight angle, $180°$, at O.
2. By Theorem 9, $\angle ACB = \dfrac{180°}{2} = 90°$.

Things to Remember

The angle at the centre is double the angle on the circle, for the same arc.
Angles in the same segment are equal.
The angle in a semicircle is a right angle.

Solve first.

P7. Draw a circle and a chord AB. Mark a point C on the major arc, a point P inside the circle and a point Q outside it, all on the same side of AB. Measure $\angle ACB$, $\angle APB$ and $\angle AQB$. Which is the largest? Which is the smallest?

Example 2. AB is a diameter of a circle with centre O, and C is a point on the circle with $\angle CAB = 35°$. Find $\angle ABC$.

FIG

Given: AB is a diameter; $\angle CAB = 35°$.
To find: $\angle ABC$.

1. $\angle ACB = 90°$ (angle in a semicircle).
2. $\angle ABC = 180° - 90° - 35° = 55°$ (angle sum of a triangle).

Answer: $\angle ABC = 55°$.

Draw a rough figure for every question in Exercise 3A that has no figure.

Exercise 3A

1. An arc of a circle subtends an angle of $70°$ at the centre. Find the angle it subtends at a point on the rest of the circle.

2. In a circle with centre O, C is a point on the major arc AB and $\angle ACB = 40°$. Find $\angle AOB$ and $\angle OAB$.

3. In a circle with centre O and radius 12 cm, $\angle AOB = 60°$. Find the length of the chord AB, and the angle that AB subtends at a point on the major arc.

4. In a circle with centre O, $\angle AOB = 100°$, and D is a point on the *minor* arc AB. Ravi writes $\angle ADB = 50°$. Find his mistake, and find $\angle ADB$.

5. AB is a diameter of a circle and C is any other point on the circle. Find $\angle ACB$ and explain your reasoning without using Theorem 9: join OC and use two isosceles triangles.

6. Find x in each figure. O is the centre.

FIG

7. A carpenter wants to check that a wooden piece has a true semicircular edge. She places the corner of a set square on the curved edge so that its two edges pass through the ends of the straight edge. She repeats this at several places. Explain the test.

8. $MNOP$ is a quadrilateral inscribed in a circle, and MN is a diameter. What can you say about $\angle MOP$ and $\angle MNP$? Explain.

9. A and B are points on a circle.

   (a) Can there be points X and Y on the circle, on the same side of AB, with $\angle AXB \neq \angle AYB$?

   (b) If X and Y are on the circle and $\angle AXB = \angle AYB$, must X and Y be on the same side of AB?

   (c) If $\angle AXB = \angle AYB$, and X and Y are not on the circle, does the circle through A, B and X also pass through Y?

10. In Class 2 we saw by construction that the circumcentre of a right-angled triangle is the midpoint of its hypotenuse. Prove it. Let $\triangle PQR$ have $\angle P = 90°$ and circumcentre O, and use Theorem 9 for the arc QR that does not contain P.

11. A chord of a circle is equal to the radius. Find the angle it subtends at a point on the major arc.